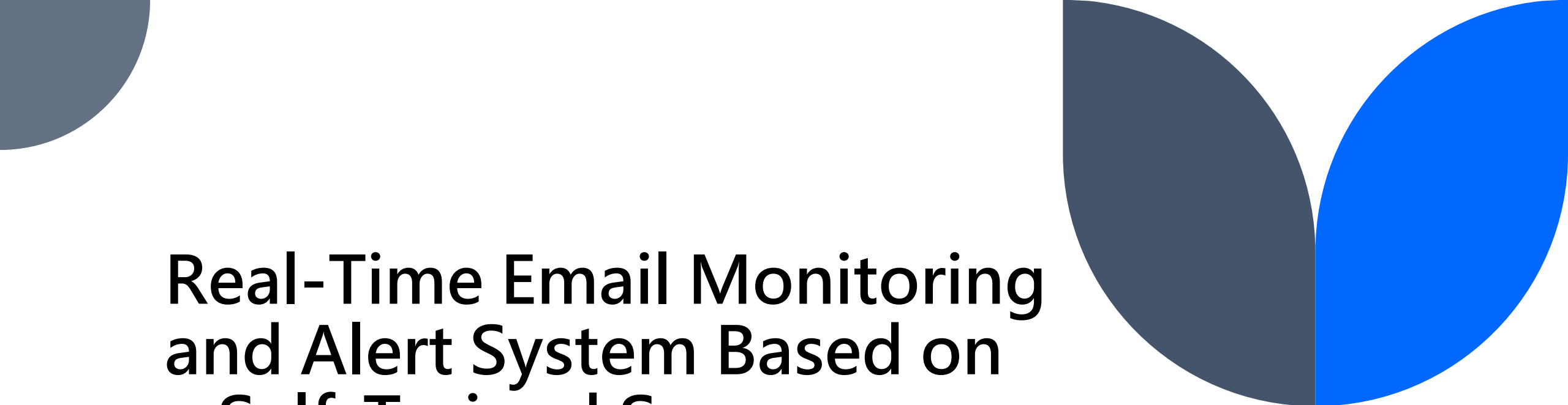
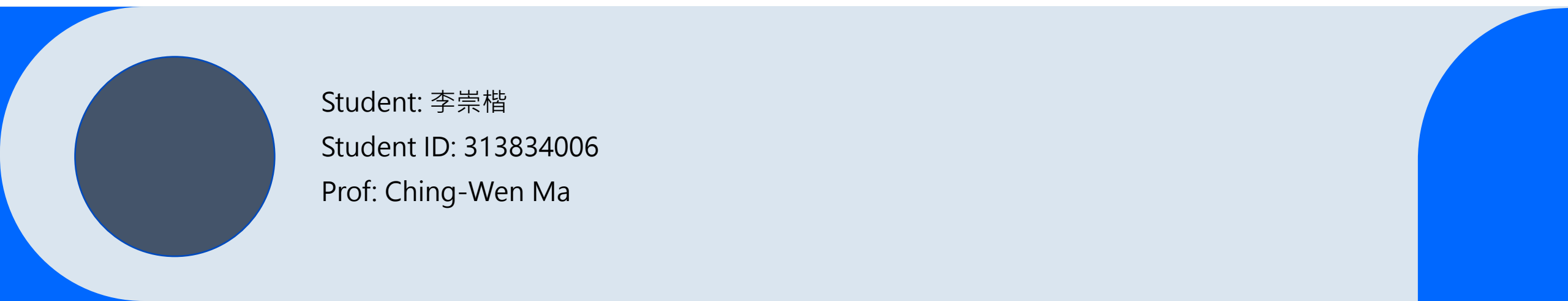




Real-Time Email Monitoring and Alert System Based on a Self-Trained Spam Detection Model



Student: 李崇楷

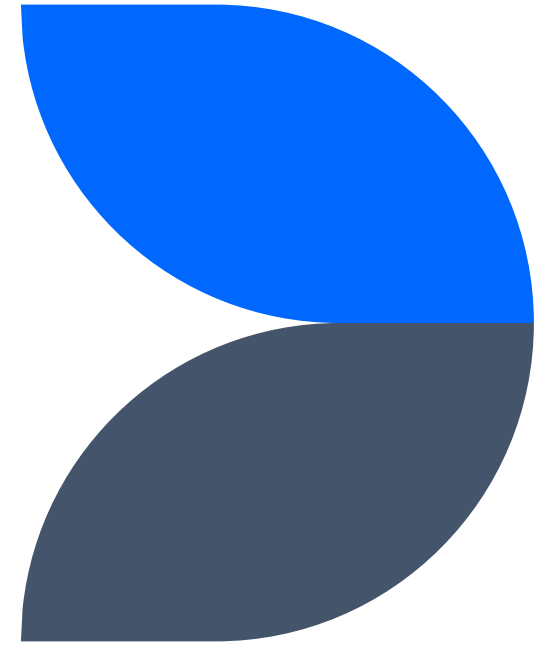
Student ID: 313834006

Prof: Ching-Wen Ma

Outline

- Introduction
- Structure
- Method
- Reference

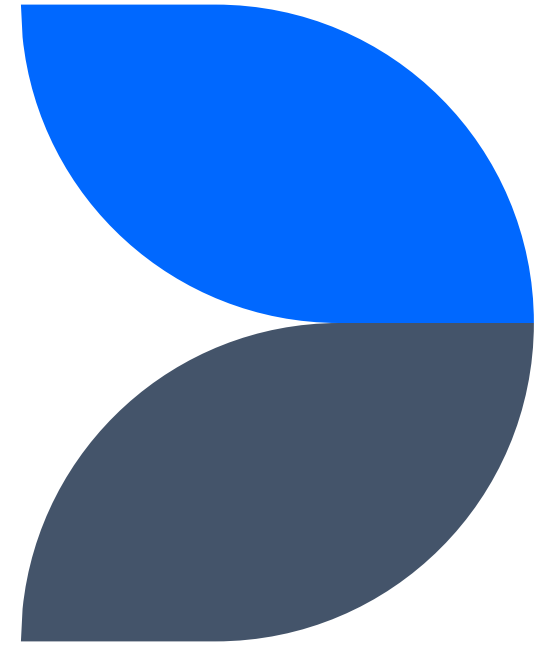
Introduction



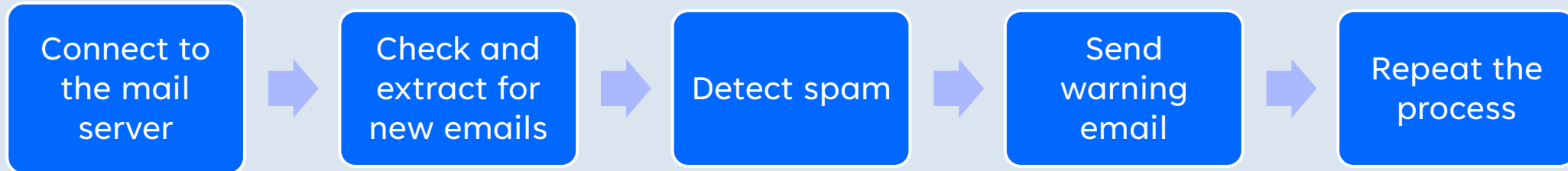
Introduction

While Gmail and other email providers offer spam detection, they are not foolproof and often fail to catch sophisticated phishing attempts. To address this, we developed an email monitoring system using a custom-trained spam detection model. By leveraging the IMAP protocol, our system fetches emails, analyzes them, and alerts users if a message is flagged as spam or phishing. This ensures enhanced protection by addressing the limitations of existing spam filters with a tailored machine-learning approach.

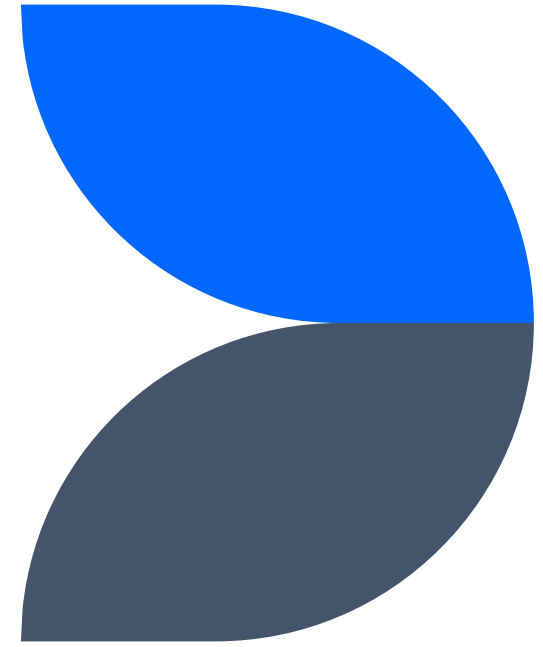
Structure



Structure



Method



Connect to the Mail Server

IMAP (Internet Message Access Protocol) is a protocol used for remotely accessing and managing emails from a mail server. It allows users to view folder structures, download email metadata or partial content (such as attachments), and perform actions (such as marking emails as read, moving, or deleting emails).

```
# Connect to Gmail's IMAP server
imap = imaplib.IMAP4_SSL("imap.gmail.com")
imap.login(username, password)
print("Login successful!")
```

```
Connecting to IMAP server...
Login successful!
```


Check and Extract for New Emails

```
# Select the mailbox folder
imap.select(folder)
print(f"Selected folder: {folder}")

# Search for all emails
status, messages = imap.search(None, "ALL")
mail_ids = messages[0].split()[-5:]
print(f"Found {len(mail_ids)} emails to process")
```

```
Found 5 emails to process
Fetching email 1/5...
Fetching email 2/5...
Fetching email 3/5...
Fetching email 4/5...
Fetching email 5/5...
Fetching emails complete.
```

Subject: 您已經繳交 Week11 (11/15) Seminar Reflection Submission 作業了

Body: 1131.639406 ->作業 ->Week11 (11/15) Seminar Reflection Submission您已經繳交了作業“Week11 (11/15) Seminar Reflection Submission” 點擊以查看您 繳交的作業。

Subject: 【陽明交通大學 校園公告系統】2024/11/15 新增訊息公告

Body: 今日新增之公告 近 3 天到期之公告 當日活動公告 今日新增之公告 到期日 主旨 公告單位 公告類別 2024/11/28 中山醫學大學附設醫院於113年12月7日至12月8日舉辦「2024大數據醫學研究國際研討會」。研究發展處企劃一組 電子

Subject: 【Amazon X NYCU】113年11月27日 16:30-18:00 Amazon X NYCU ECE X CTLD Tech Talk 2024

Body: Amazon ◆ NYCU ECE ◆ CTLD Tech Talk 2024 報名: <https://forms.gle/ygphhbWuJBwxeBKF9> 活動日期: 113年11月27日 活動時間: 16:30-18:00 活動地點: 交大校區工程四館116教室 ☀ 講師: Amazon團隊 ★★★★★講座大綱

Subject: 服務學習中心【國際志工成果靜態展】活動開跑啦~歡迎大家來觀展! Service-Learning Center 【International Volunteer Achievement Static Exhibition】is starting!

Body: 2024 國立陽明交通大學印度、東南亞國際志工團成果靜態展啟動囉!【我們去了哪些地方?】【今年志工團們做了些什麼?】快揪好朋友們一起來看展,一口氣帶你們了解~【活動資訊】展期:2024/11/1-2024/11/29地點:浩然圖書館2F

Subject: 安全性快訊

Body: 用於登入帳戶的應用程式密碼已產生 kevin611116.ai13@nycu.edu.tw 假如您並未替「IMAP」產生這個密碼,表示可能有人盜用了您的帳戶。請立即檢查帳戶並採取必要保護措施。查看活動您也可以前往以下網址查看帳戶安全相關活動

Data Preprocessing

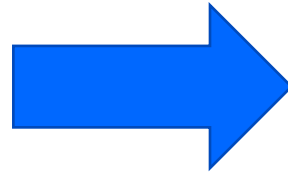
Method	Advantages	Disadvantages
Word2idx	- Simple and fast - Memory-efficient	- No semantic understanding - Fixed vocabulary
Hashing Vectorizer	- Efficient for large datasets - Fixed size vectors	- Potential hash collisions - No semantic relationships
Word2Vec	- Captures semantic relationships - Pre-trained models available	- Requires large training data - Context-independent
BERT-base-uncased	- Context-aware - Handles subwords - Highly versatile	- Computationally intensive - High memory usage

Data Preprocessing- Word2idx

Category		Message
ham spam	87% 13%	5157 unique values
ham		Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got a...
ham		Ok lar... Joking wif u oni...
spam		Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entr...

Kaggle Dataset

Text Vectorization



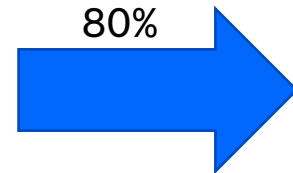
Vectorized Texts:

```
[[0 0 0 0 1 0 0 0 0 1]
 [1 0 1 0 0 1 0 0 1 0]
 [0 1 0 1 0 0 1 1 0 0]]
```

Encoded Labels:

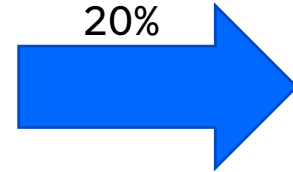
```
[0 1 0]
```

80%



Train

20%



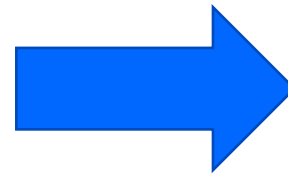
Val

Data Preprocessing- Hashing Vectorizer

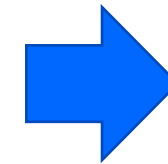
Category		Message
ham spam	87% 13%	5157 unique values
ham		Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got a...
ham		Ok lar... Joking wif u oni...
spam		Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entr...

Kaggle Dataset

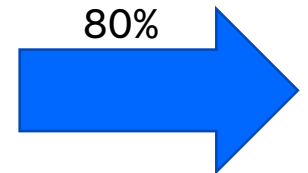
Hash function



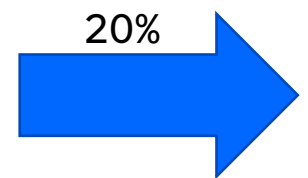
[hash(Text).....]



Hash Vector



Train



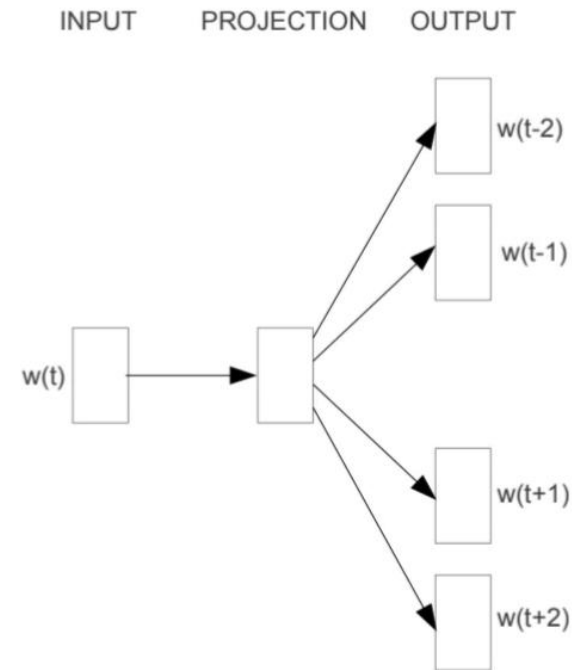
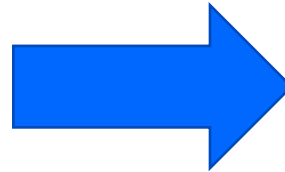
Val

Data Preprocessing- Word2Vec

Category		Message
ham spam	87% 13%	5157 unique values
ham		Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got a...
ham		Ok lar... Joking wif u oni...
spam		Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entr...

Kaggle Dataset

Word2Vec

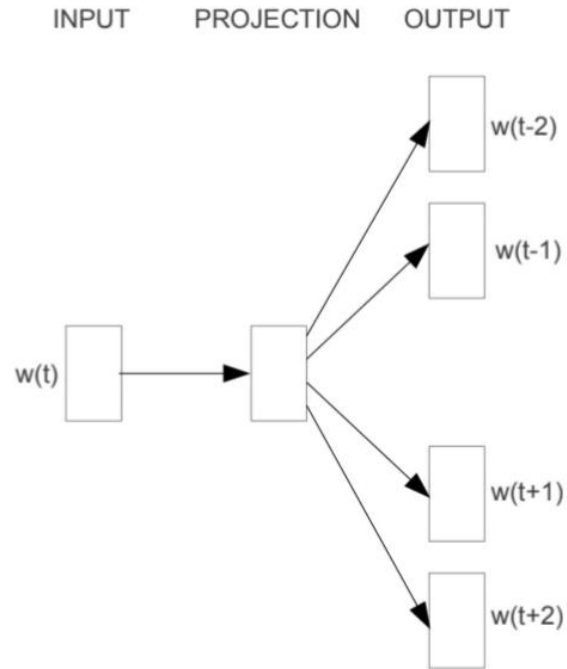
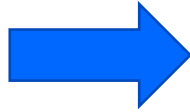


80% → Train

20% → Val

Skip gram

Word2Vec
 $[w_1, w_2, w_3, \dots, w_n]$



$$P(w_{t+j}|w_t) = \frac{\exp(\text{vec}(w_{t+j})^\top \text{vec}(w_t))}{\sum_{w \in V} \exp(\text{vec}(w)^\top \text{vec}(w_t))}$$

$$\text{NLL} = - \sum \log P(w_{t+j} | w_t)$$

Data Preprocessing- Bert-base-uncased

Category		Message
ham spam	87% 13%	5157 unique values
ham		Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got a...
ham		Ok lar... Joking wif u oni...
spam		Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entr...

Kaggle Dataset

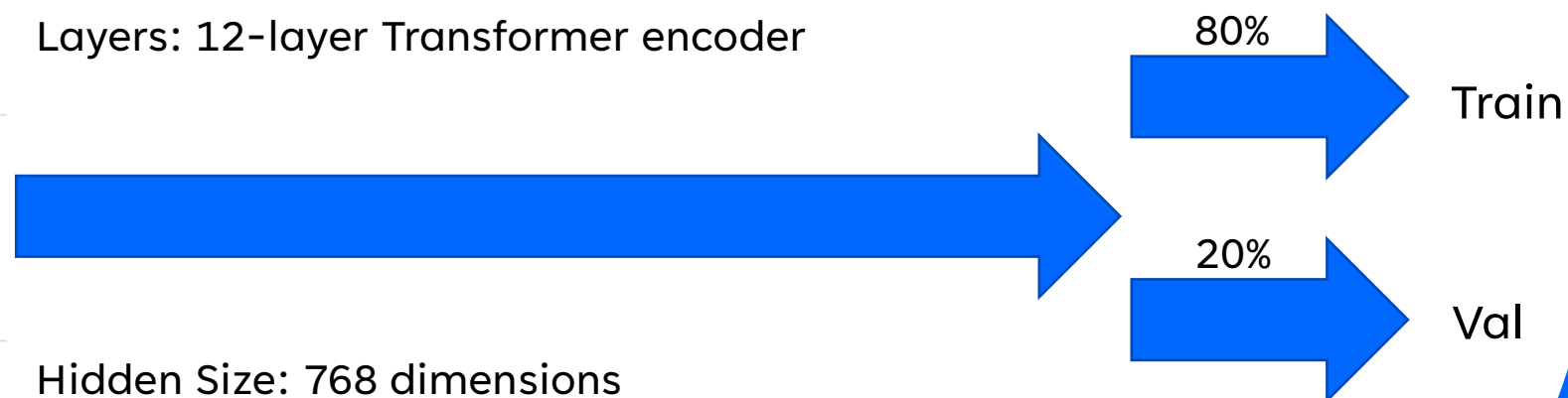
Bert-base-uncased:

Layers: 12-layer Transformer encoder

Hidden Size: 768 dimensions

Attention Heads: 12

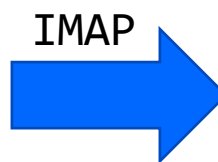
Total Parameters: Approximately 110 million



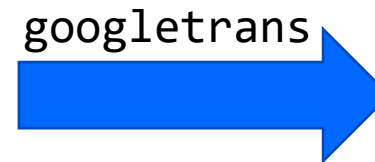
Data Preprocessing -Test



mail



Subject(Chinese)
body(Chinese)



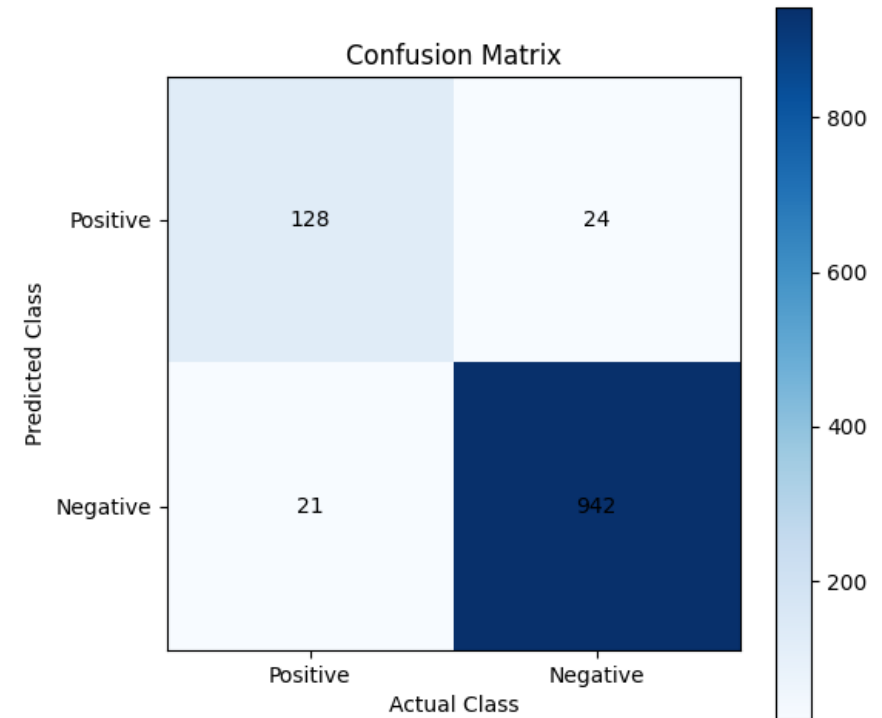
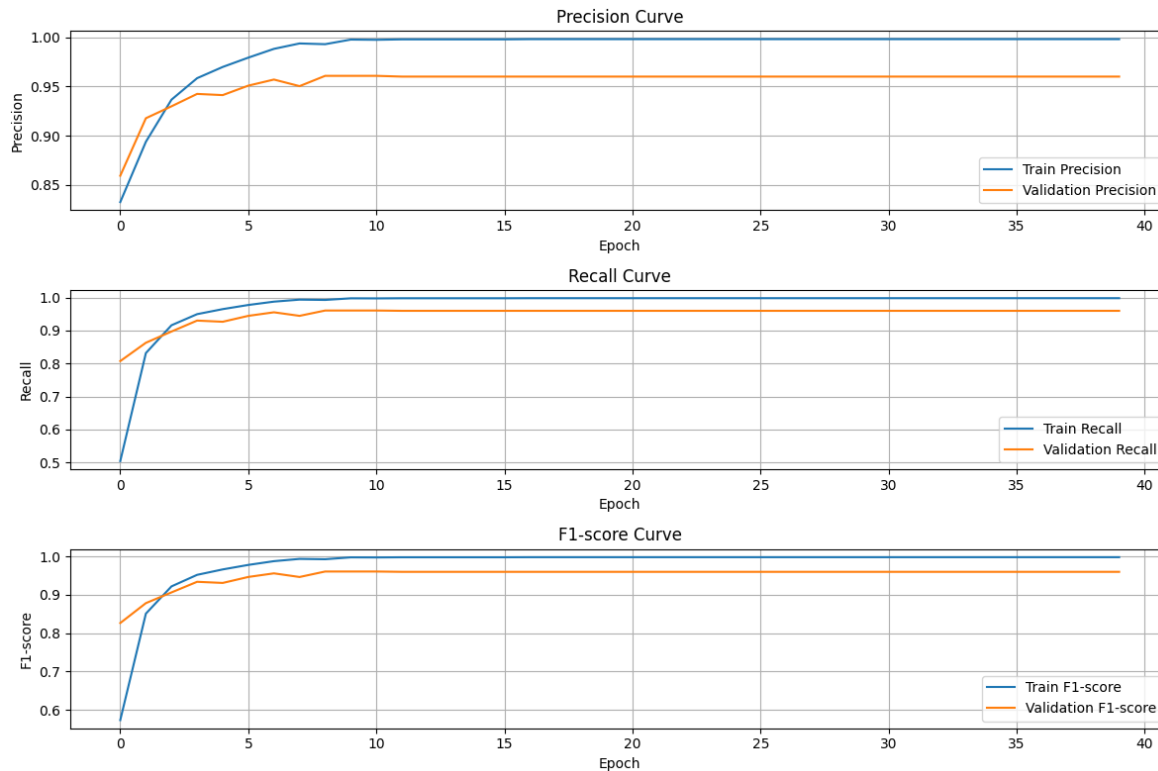
Subject(English)
body(English)

Text Vectorization

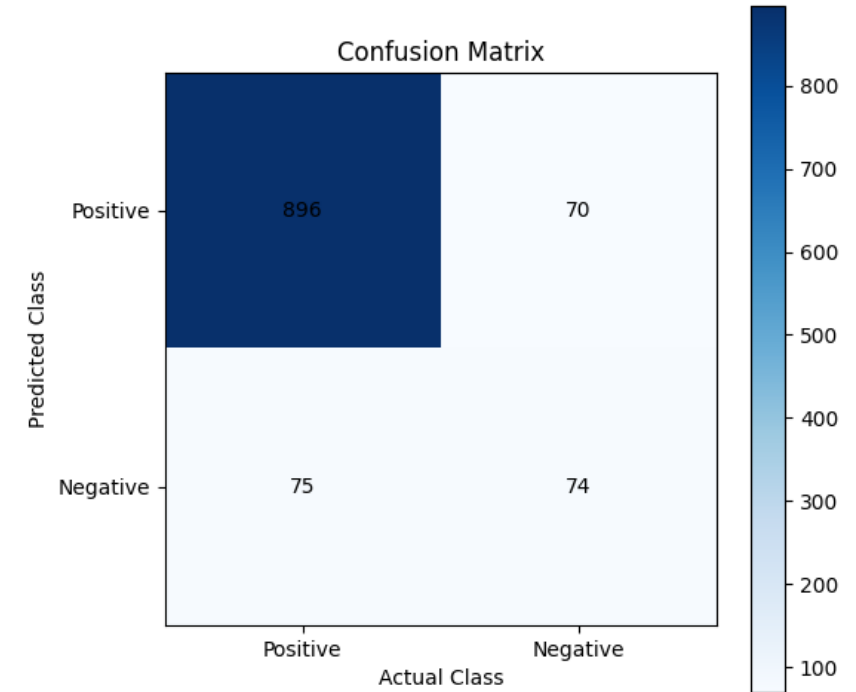
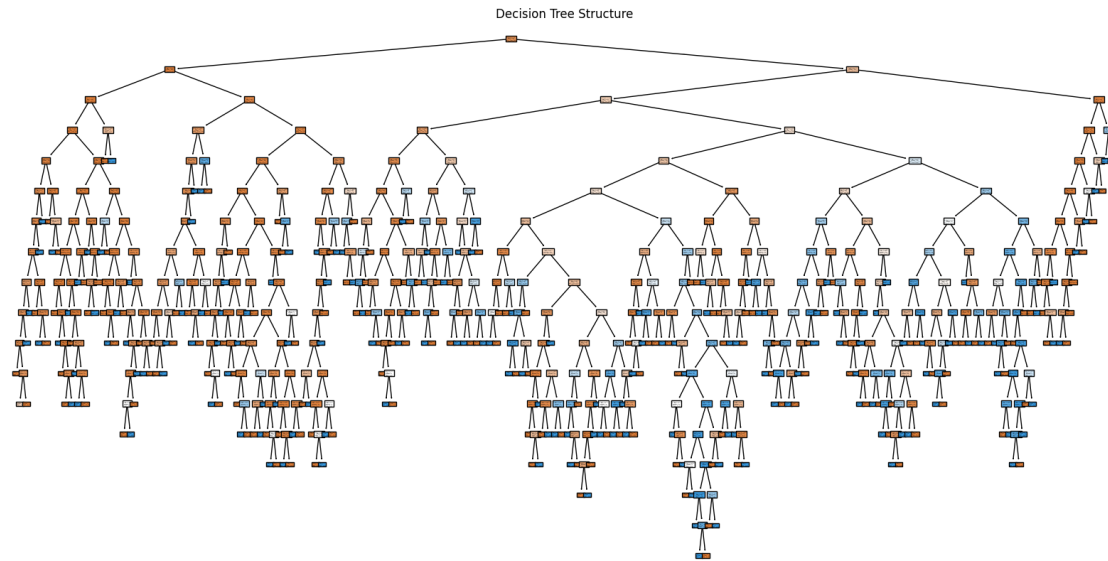


Test

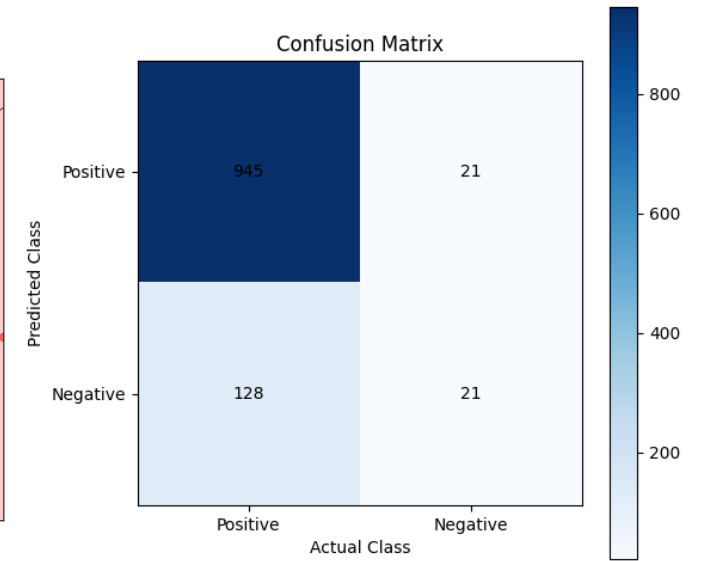
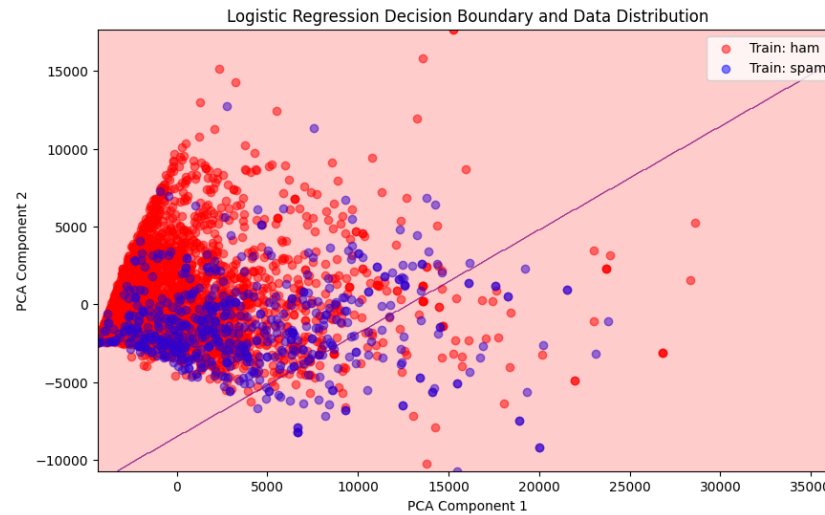
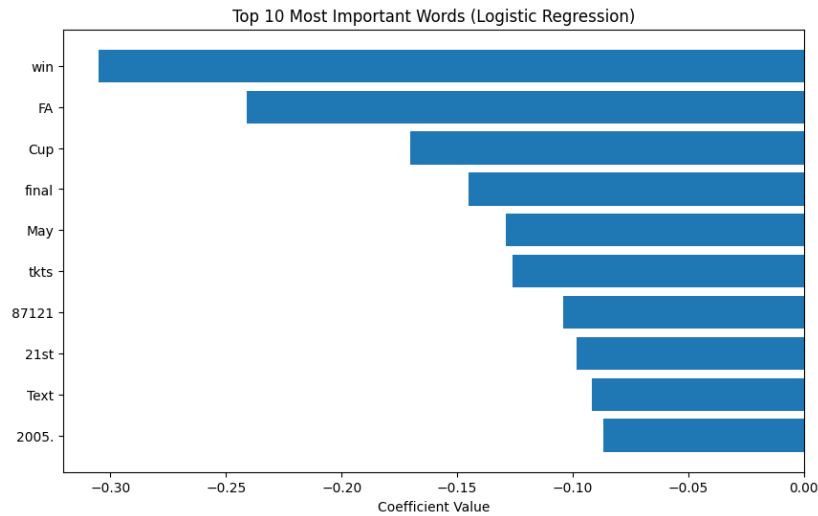
Model -LSTM



Model – Decision Tree



Model – Logistic Regression



Result- Word2idx

Train	LSTM	Decision Tree	Logistic Regression
Precision	99.80%	99.96%	85.46%
Recall	99.80%	99.96%	87.73%
F1-score	99.80%	99.96%	85.50%

Val	LSTM	Decision Tree	Logistic Regression
Precision	96.00%	86.81%	82.98%
Recall	95.96%	87.00%	86.64%
F1-score	95.98%	86.90%	83.24%

Result- Hashing Vectorizer

Train	LSTM	Decision Tree	Logistic Regression
Precision	99.87%	100%	97.41%
Recall	99.87%	100%	97.37%
F1-score	99.87%	100%	97.27%

Val	LSTM	Decision Tree	Logistic Regression
Precision	98.48%	96.57%	97.19%
Recall	98.48%	96.59%	97.13%
F1-score	98.48%	96.58%	96.99%

Result- Word2Vec

Train	LSTM	Decision Tree	Logistic Regression
Precision	92.40%	100%	96.26%
Recall	89.25%	100%	96.34%
F1-score	90.19%	100%	96.21%

Val	LSTM	Decision Tree	Logistic Regression
Precision	91.68%	94.01%	95.53%
Recall	87.80%	93.90%	95.61%
F1-score	88.97%	93.95%	95.41%

Result- BERT-base-uncased

Train	LSTM	Decision Tree	Logistic Regression
Precision	99.89%	100%	99.93%
Recall	99.89%	100%	99.93%
F1-score	99.89%	100%	99.93%

Val	LSTM	Decision Tree	Logistic Regression
Precision	99.46%	96.19%	99.11%
Recall	99.46%	96.23%	99.10%
F1-score	99.46%	96.21%	99.11%

Send warning email

SMTP (Simple Mail Transfer Protocol) is a communication protocol used for transmitting emails over the network. Its primary purposes are sending emails and forwarding emails, and it can support multiple formats like plain text, HTML, and emails with attachments.



您可能收到可疑的郵件或訊息，請務必提高警覺！詐騙郵件通常會：

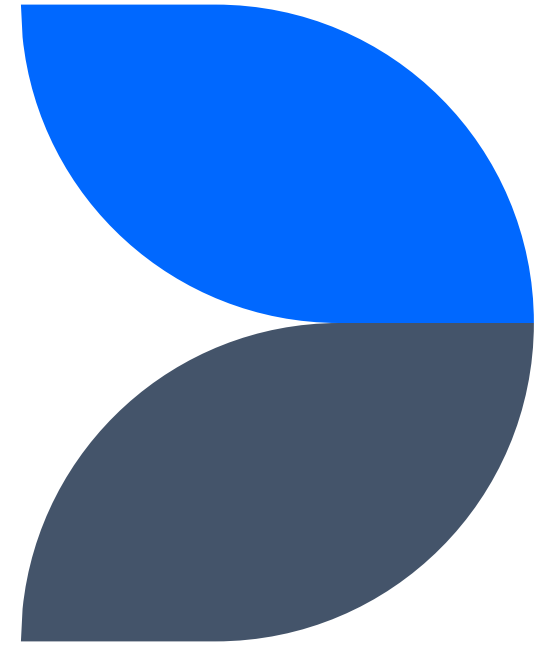
- 要求您點擊可疑連結。
- 要求提供您的銀行帳號、密碼或其他個人資料。
- 包含讓您感到緊急的內容，例如：「帳戶即將被凍結」或「贏得大獎」。

請勿點擊任何不明連結，並仔細核對郵件發件人地址。如有疑慮，請聯繫官方客服。



保護您的安全是我們的責任，請時刻保持警惕。

Conclusion



Conclusion

In the end, I chose to use an LSTM combined with BERT-base-uncased as my word tokenizer to implement a model capable of recognizing spam. Through two different protocols, the system can promptly capture users' emails and notify them in real-time. In the future, I hope to integrate my own dataset for further training.

謝謝大家

